INCH-POUND
MIL-R-70561B (AR}
January 1992
SUPERSEDING
(See 6.6)

MILiTARY SPECIFICATION

RECEIVER-TRANSMITTER, RADAR SET RT-860A/VPS-2A: 9361152

This specification is approved for use by the U. S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements and tests for the Receiver Transmitter, Radar Set RT-860A/VPS-2A, hereinafter referred to as the receiver-transmitter, which is part of Radar Set AN/VPS-2A. The radar receiver-transmitter generates the high power transmit pulse for radiation by the antenna. In addition, the receivertransmitter receives the target ethos and sends them to the receiver. The receiver-transmitter contains a Built-In-Test circuit assembly for **signal** monitoring and weapon system fault isolation (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-F-13926 - Fire Control Materiel, Manufacture and Inspection, General Specification for
MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance and Disposition of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should b e addressed to: Commander, U.S. Army ARDEC, ATtn:SMCAR-BAc.S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed standardization Document improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1285

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.
Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:41Z
Check the source to verify that this is the current version before use.

MIL-T-50320 - Transmit Gate Assembly: 10549202
MIL-P-50321 - Klystron Power Supply Assembly: 11834449
MIL-P-50323 - Klystron Beam Power Supply Assembly: 11834450
MIL-P-50324 - Pulse Shaper Assembly: 10548801
MIL-R-70560 - PIVADS Range Only Radar Set AN/VPS-2A: 9360797
MIL-C-70574 - Circuit Board, Built-In-Test, 2A8 Unit 2 Assembly (PIVADS): 9361010

STANDARDS

MILITARY

MIL-STD-109 - **Quality** Assurance Terms and Definitions
MIL-STD-810 - Environmental Test Methods

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS (See 6.4)

U. S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9361152 - Receiver-Transmitter, Radar RT-860A/VPS-2A
9361217 - Remote BIT Tester Assembly
10548242 - Relay Time Delay, SPNO
12950877 - Requirements for Soldering Electrical and Electronic Assemblies

SPECIAL PACKAGING INSTRUCTION

PS9361152 - Packaging of Receiver-Transmitter, Radar RT-860A/VPS-2A

(Copies of drawings, publications **and** other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:41Z
Check the source to verify that this is the current version before use.

2.2 Order of precedence. In the event of conflict between the text of this document and the references cited herein the text of this document shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 9361152, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the receiver-transmitter shall meet the requirements of MIL-F-13926 as follows:

a. Dimensions and tolerances.
b. Inorganic protective surface finishings.
Part identification and marking.
d. Workmanship.

3.2.1 Subassemblies. The sub-assemblies of the receiver-transmitter shall have been inspected and accepted in accordance with respective drawings and specifications prior to their installation in the receiver-transmitter. The following subassemblies are covered by the indicated specifications:

Unit No.	Subassembly	Specification
2AI	Microwave Chassis Assembly, 93613717	MIL-M-70705
2A2	Klystron Power Supply Assembly, 11834449	MIL-P-50321
2A5	Transmit Gate Assembly, 10549202	MIL-T-50320
2A6	Klystron Beam Power Supply Assembly, 11834450	MIL-P-50323
2A7	Pulse Shaper Assembly, 10548801	MIL-P-50324
2A8	Circuit Board, Built-In-Test, 2A8 Unit 2 Assembly (PIVADS), 9361010	MIL-C-70574

Source: https://assist.dla.mil/ Downloaded: 2026-10-23T03:41Z
Check the source to verify that this is the current version before use.

3.3 Environmental service conditions.

3.3.1 Vibration. The receiver-transmitter shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to the equivalent of 6000 miles of track-laying vehicle operational vibration as defined by MIL-STD-810 narrowband random-on-random data for equipment installed on the deck of the **M113** armored personnel carrier.

3.3.2 Shock. The receiver-transmitter shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to a total of 18 impulse shocks consisting of three shock impulses of 15 a's + 10%. 11 milliseconds + 10%, half-sine wave applied in each direction-of 3 mutually perpendiclar axes.

3.3.3 Altitude Operating. The receiver-transmitter shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches of Hg).

3.3.4 Temperature.

3.3.4.1 Operating. The receiver-transmitter shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between -40°F and +155°F.

3.3.4.2 Non-operating (storage). The receiver-transmitter shall not suffer permanent damage when subjected to the constant temperature extremes of -60°F for a period of 24 hours and +160° for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (non-operating). The receiver-transmitter shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:41Z
Check the source to verify that this is the current version before use.

3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram of the receiver-transmitter.

3.5 Performance. The unit under test shall be allowed 2 minutes warm-up time prior to testing, except that 15 minutes is permitted for the low temperature tests of 4.7.1.4.2. All test equipment shall be allowed 15 minutes warm-up time prior to testing.

3.5.1 Interlock switch. When the receiver-transmitter cover is open, an open circuit shall exist between 2J1-F and 2J1-H. When the receiver-transmitter cover is closed continuity shall exist between 2JI-F and 2J1-H. A manual override feature shall be included in the interlock switch to provide continuity between 2J1-F and 2J1-H when the receiver-transmitter cover is open and the interlock switch plunger is in the manual override position.

3.5.2 Clutter lock-on switch. The clutter lock-on switch shall provide continuity between 2J1-b and 2J1-Z when in the normal position. There shall not be continuity between 2J1-b ***and*** 2TB1-1. Alternatively, when this clutter lock-on switch is in the test position, there shall be continuity between 2J1-b and ZTB1-I. There shall be no continuity between 2J1-b and 2J1-Z.

3.5.3 overload reset switch. When the overload reset switch is operated to either of its reset positions from the springloaded center off position, the switch shall provide continuity between 2TB1-1 and 2P18-2.

3.5.4 Indicator lamps.

3.5.4.1 Modulator overload indicator. The modulator overload lamp, 2DS1, shall illuminate when 28 volts dc (Vdc) is applied between 2P18-3 and 2TB1-2.

3.5.4.2 High voltage Power supply overload indicator. The high voltage power supply overload lamp, **2DS2**, shall illuminate when 28 Vdc is applied between 2P18-1 and 2TB1-2.

3.5.4.3 Power indicator lamp. when 24 Vdc is applied between 2J1-G and 2J1-H and the door interlock switch is closed, the power indicator lamp 2DS-3 shall illuminate.

3.5.5 Standby operation.

3.5.5.1 Blower operation. The blower, 2B1, shall operate and provide an intake airflow.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:41Z
Check the source to verify that this is the current version before use.

3.5.5.2 Gated stalo output. With the modulator trigger applied to 2J2-F, a gated 60 megahertz (MHz) stalo signal shall be present at 2J2-G and-shall exhibit the following parameters:

Amplitude: 0.1 (+0.05, -0.00) V peak-to-peak minimum.
width: **(50** percent points) 280 + 10 nanoseconds (ns).
PRF : 20 KHz.

3.5.5.3 Pulse shaper output. The 2A7 pulse shaper output shall appear at 2P23 with the following parameters:

Amplitude: +15.0 V peak minimum.
Width: *Minimum: 0.60 microseconds @ 60°F to 90°F.
**Maximum: 1.00 microseconds @ 60°F to 90°F'.

***Minimum:** **0.50** microseconds @ -40°F.
**Maximum: 1.15 microseconds @ +155°F.

3.5.5.4 Overall gain. The overall gain of the receiver-transmitter, measured from 2J3 to 2J5-D with 0.0 Vdc at 2J5-E, shall be 17 decibels (DB) minimum, 22 DB maximum.

3.5.5.5 Automatic Gain Control (AGC) action. The gain of the receiver-transmitter shall be reduced by a minimum of 20 DB when the dc voltage of 2J5-E is changed from 0.0 Vdc to +6.0 Vdc. The gain versus AGC voltage curve shall be monotonic.

3.5.5.6 Noise figure. The noise figure for the receiver-transmitter shall be 12 DB maximum on all six channels.

3.5.5.7 switch drive pulse. The switch drive pulse at 2A1A7J1 shall exhibit the following parameters:

Negative amplitude: 22 Vdc minimum.
Positive reference: +0.7 vdc minimum, +1.1 Vdc maximum
Width: (50 percent points) 2.48 microseconds minimum, 2.62 microseconds maximum.

3.5.5.8 Diode switch isolation. With +1.0 Vdc applied to 2A1A7J1, the insertion loss shall be 0.75 DB maximum. When the dc level at 2A1A7J1 is changed from +1.0 Vdc to -33.0 Vdc, the insertion loss of the diode switch shall increase **a** minimum of 45 DB .

3.5.6 Radiate mode.

3.5.6.1 Average output power- The average output power, measured at 2J3, shall be 4.26 watts (W), minimum.

Source: https://assist.dla.mil -- Downloaded: 2026l0-23T03:41Z
Check the source to verify that this is the current version before use.

3.5.6.2 Transmit pulse timing. The transmit pulse timing shall be such that the delay between the Klystron modulator pulse edge at the start of the flat and the center of the gated 60 MHz RF (COS^2) pulse shall be 0.3 microseconds minimum, 0.32 microseconds maximum.

3.5.6.3 Transmit Pulse envelOPe. The transmit pulse envelope, when detected and observed on an oscilloscope, shall have a cosine squared (COS^2) characteristic shape and shall be 280 to ± 2018 ns wide at the 50 percent amplitude points.

3.5.6.4 Transmit frequency. The receiver-transmitter unit shall be capable of operation on six discrete frequencies by selecting one of six oscillator plug-in modules. The frequency of the transmitted signal shall be 9205 MHz ± 10 MHz using Channel 1 oscillator module. Using the Channel 1 measurement as a reference, the Channel 2 through 6 output frequencies shall be as follows:

Channel 2	Reference + 8 ± 2 MHz
Channel 3	Reference + 16 ± 2 MHz
Channel 4	Reference + 24 ± 2 MHz
Channel 5	Reference + 32 ± 2 MHz
Channel 6	Reference + 40 ± 2 MHz

3.5.7 Subclutter visibility test. The receiver-trmsmitter after having met the requirements of 3.5.1 through 3.5.6 shall meet the subclutter visibility (SCV) requirements as specified in MIL-R-70560 when substituted into Radar Set AN/VPS-2A. Radar Set AN/VPS-2A and facilities shall be as specified in MIL-R-70560.

3.5.8 Remote BIT assembly. The receiver-transmitter shall contain a remote Built-In-Test (BIT) circuit card assembly (2A8) to monitor control and operation of the receiver-transmitter and to send data to the fire control processor when the system is in radar and BIT modes. The BIT assembly performance shall be as specified in MIL-C-70574.

3.6 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.7 Soldering. All soldering shall be in accordance with the general requirements of Note 2 of Drawing 12950877.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance-of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:41Z
Check the source to verify that this is the current version before use.

requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Reference shall be made to MIL-STD-109 to define **quality assurance terms used herein.**

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c* Special sampling inspection (see 4.5).

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

	Temperature	+50° to +90°F
b.	Relative humidity	90% RH max.
c.	Pressure	28 to 32 inches of Hg

4.3 First article inspection.

4.3.1 First article submission. The contractor shall submit a first article sample as designated by the contracting officer consisting of three (3) receiver-transmitters for inspection in accordance with 4.3.2.

4.3.2 Inspections to be performed. First article assemblies, components and test specimens may be subjected by the Government to any or all Of the examinations and tests-specified in Table I (Functional Test) and Table II (Special Sampling) herein and to any or all requirements Of the applicable drawings.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:41Z
Check the source to verify that this is the current version before use.

4. 3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component or test specimen to comply with any of the stated requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term 'inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions, and complies with the provisions for submission of product. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies.

The contractor's quality program or detailed inspection plan shall provide assurance of compliance of all characteristics with the applicable drawings and specifications utilizing as a minimum the conformance criteria specified in this document and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAP's, the applicable quality assurance provisions of MIL-F-13926 shall apply.

b. Alternative inspection procedures, methods or equipment, such as statistical process control, tool control, other types of sampling procedures, etc. , may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the procuring contracting officer for evaluation and approval by the Government. When required, the contractor shall demonstrate that the effectiveness of the proposed alternative(s) is equal to or better than the specified quality assurance provisions herein. In cases of dispute as to whether the contractor's proposed alternative(s) provide equal assurance, the provisions of this specification shall

Source: https://assist.dla.mil - Downloaded: 2026-10-23T03:41Z
Check the source to verify that this is the current version before use.

apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program plan or detailed inspection system, as applicable.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examinations and tests related to Section 3 shall be performed as specified in 4.4.2.3 and 4.5.

4.4.2.3 Functional tests. The tests in Table I shall be performed on a 100% basis.

TABLE I. Functional tests (100 percent inspection).

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
101. Electrical	3.4	4.7.2
102. Performance	3.5	4.7.3
Minor:	None	N/A

4.5 Special sampling. One receiver-transmitter out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests of Table I and shall then meet the tests in Table II. Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.

TABLE II. Special sampling.

CHRACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
103. Vibration	3.3.1	4.7.1.1
104. Shock	3.3.2	4.7.1.2
105. Altitude	3.3*3	4.7.1.3
106. Temperature	3.3.4	4.7.1.4
107 . Humidity	3*3.5	4.7.1.5
Minor:	None	*N/A*

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:41Z
Check the source to verify that this is the current version before use.

4.5.1 Failure of sample. Should any one item of a special Sampling fail to meet any of the specified test requirements, acceptance of the Product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications *or QAPs* forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and 'Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled **as** to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2 the test equipment listed in Table III (Test Equipment list) shall be used in the performance of the applicable test methods and procedures as specified in 4.7.

Source: https://assist.dla.mil. Downloaded: 2026-10-23T03:41Z
Check the source to verify that this is the current version before use.

TABLE III. Test equipment list.

ITEM NUMBER	DESCRIPTION
1. X-band RF load	Band width: 9150 to 9250 MHZ minimum. Average power rating: 10 W minimum. VSWR : 1.0:1.1 maximum.
2. X-band detector	Bandwidth: 9150 tO 9250 MHZ minimum. Average power rating: 10 mW minimum. VW-R : 1.0:1.8 maximum with 100 ohm load and 0.5 mA crystal current.
3. X-band coupler (2 required)	Bandwidth: 9150 to 9250 MHZ minimum. Peak power rating: 2000 W minimum. Coupling: 40 ± 0.9 DB. Mainline VSWR: 1.0:1.1 maximum. Secondary line VSWR : 1.0:1.2 maximum.
4. X-band signal generator	Frequency range: 9150 MHz to 9250 MHz minimum. Frequency stability: With temperature - less than 0.006 percent/°C change in ambient temperature; with line voltage - less than 0.02 percent change for line voltage variation of ± 10 percent. Residual FM: Less than 15 kHz peak. output range: 0.224 V to 0.1 microvolt into a 50-ohm load. Output calibration: O DBM to -127 DBM. Source impedance: 50 ohms nominal.
5. X-band frequency meter	Type: Direct reading resonant cavity. Frequency range: 9150 to 9250 MHz minimum. Overall accuracy: 0.08 percent.
6. Thermistor mount	Frequency range: 9150 to 9250 MHZ minimum. Maximum average input power: 10 mW minimum.
7. Power meter	Frequency range: 9150 to 9250 MHZ minimum. Power range: 1 mW full scale. Accuracy: ± 1 percent of full scale. Type: Thermistor sensing element.

Source: https://assist.dla.mil -- Downloaded: 202610-23T03:41Z
Check the source to verify that this is the current version before use.

TABLE III. Test equipment list. (continued)

ITEM NUMBER	DESCRIPTION
8. I-f amplifier	Frequency: 60 MHZ. 3 DB bandwidth: 10 MHz minimum. Gain: 20 DB. Maximum input: Not less than 0 DBM with a maximum distortion of 5 percent.
9. Noise figure meter	Type: Automatic readout with argon tube waveguide noise source and 60 MHz integral i-f amplifier. Scale: Not less than 20 DB full scale; 12 DB calibration point shall be not less than half scale.
10* Waveguide noise source	Type: Argon-filled gas-discharge tube in standard waveguide housing. Output connections: UG-39/U. Excess noise rating: 15.75 ± 0.5 DB.
11. Oscilloscope	Vertical display: Dual trace, dc to 75 MHz (approximately 3 DB down) at 10 mV/cm minimum. Horizontal sweep: Calibrated time base 50 ns/cm to 1 s/cm minimum. Delayed sweep: 2 microsecs to 50 microsecs, continuously variable, minimum.
12. power supply	Type: Laboratory supply. Minimum requirements: 0 to 30 Vdc. Load regulation: 0.02 percent. Load current capability: 15 A. Ripple and noise: 1 mV rms or 5 mV peak-to-peak maximum from dc to 20 MHz .
13. power Supply	Type: Laboratory supply. Minimum requirements: 0 to 30 Vdc. Load regulation: 0.02 percent. Load current capability: 5 A. Ripple and noise: 1 mV rms or 5 mV peak-to-peak maximum from dc to 20 MHz .

Source: https://assist.dla.mil Downloaded: 2026-10-23T03:41Z
Check the source to verify that this is the current version before use.

TABLE III . Test equipment list. (continued)

ITEM NUMBER		DESCRIPTION
14.	Power supply	Type: Laboratory supply, independent dual voltage source. Minimum requirements: O to 20 vdc. Load regulation: 0.02 percent. Load current capability: 100 ma. Ripple and noise: l mV rms or 5 mV peak-to-peak maximum for dc to 20 MHz .
15.	Power supply	Type: Laboratory supply. Minimum requirements: O to 40 Vdc. Load regulation: 0.02 percent. Load current capability: 100 roll. Ripple and noise: 1 mV rms or 5 mV peak-to-peak maximum for dc to 20 MHz .
16.	RF voltmeter	Ranges: 3 mV to 300 mV in four ranges. Frequency range: 1 MHz to 100 MHz minimum. Accuracy: ± 3 percent of full scale.
17.	Digital voltmeter	Display: Four digit readout. Accuracy: 0.01 percent. Range: 100 Vdc.
18.	Pulse generator	pulse amplitude: +15 V peak. pulse width: 0.6 ± 0.1 US. Rise time: 100 ns maximum.
19.	Test fixture	See Figure 1 and Table IV.
20.	RF attenuator	Frequency range: Dc to 100 MHZ. Attenuation: 0 t0 30 DB in l DB steps.
21.	Remote BIT tester assembly	TS-3211/TSM-100B, NSN 4931-01-212-2436, Drawing 9261217, or equivalent.
22.	Cable assembly	NSN 4933-01-210-9301, Drawing 9361224, or equivalent.

Source: https://assist.dla.mil -- Downloaded: 2026l0-23T03:41Z
Check the source to verify that this is the current version before use.

TABLE IV. Test fixture parts list.

DESIGNATION	PART	DESCRIPTION
DS1, DS2	Lamp	No. 313
El through E12	Binding post	H. H. Smith, 257
Jl, J2, J3	Connector, BNC	UG-625B/U
Kl, K3	Relay	Potter and Brumfield, KA1lDG
K2	Relay, time delay	10548242
P1	Connector	Bendix, PT06A20-26S(SR), or equivalent
P2	Connector	Bendix, PT06SE22-78S(SR), or equivalent
P5	Connector	Bendix, PT06SE22-78SW(SR), or equivalent
R1	Resistor, variable	5000 ohms ± 10 percent, 2 W
R2	Resistor, fixed	18 Kohms ± 10 percent, 1/2 W
Sl, S3	Switch, toggle	SPST
S2	Switch, toggle	4PST
TP1, TP2	Jack	H. H. Smith 1501, or equivalent

Source: https://assist.dla.mil. Downloaded: 2026-10-23T03:41Z
Check the source to verify that this is the current version before use.

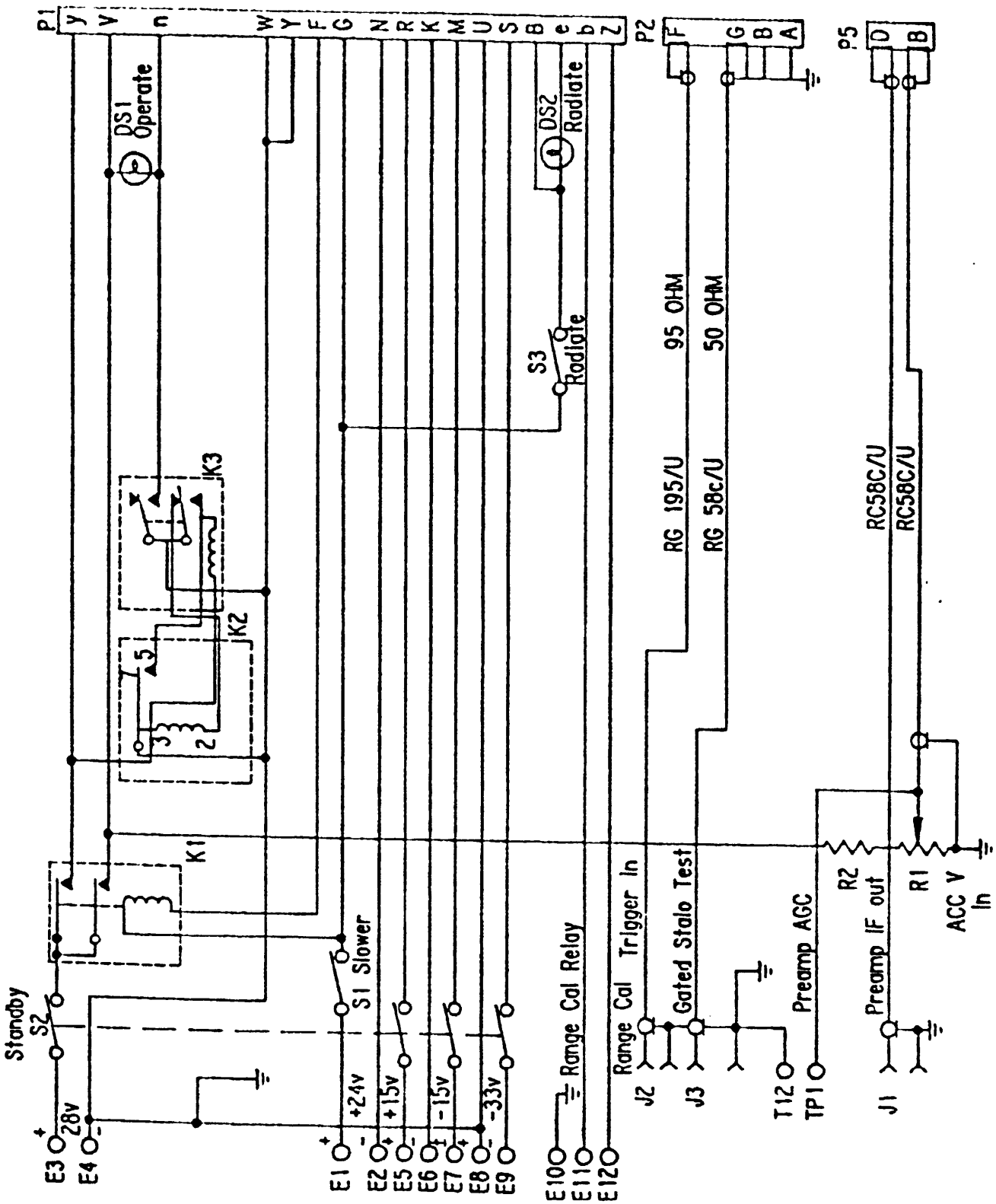


FIGURE 1. Test harness

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:41Z
Check the source to verify that this is the current version before use.

4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810, except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4.7.1.1 Vibration. The test shall be performed as specified in Method 514, Category 8. Procedure I, of MIL-STD-810 and paragraph 3.3.1 herein. In Step 1 of Procedure I only steps 4 through 8 of paragraph 11-2.1 apply. The receiver-transmitter shall be attached to the vibration generator directly or with a fixture and securely held by its normal means of attachment. After vibration, the receiver-transmitter shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.

4.7.1.2 Shock. This test shall be performed as specified in Procedure I, Section II of Method 516 of MIL-STD-810 except that:

a. The post test functional check (Step 5 of Procedure I) need not be conducted until completion of all shock impulses.

b. The shock pulses shall be 15 g's ± 10% amplitude and 11 ms ± 10% duration.

c. The maximum g shall occur at 5.5 milliseconds ± 10%.

d. The shock measuring system shall have a flat frequency response within ± lo%-from 20 to 2,000 hertz.

After the application of all the shock impulses, the receiver-transmitter shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 Altitude (operating and non-operating) This test shall be performed as specified in Procedures II Method 500, of MIL-STD-810. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.

4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 501 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:41Z
Check the source to verify that this is the current version before use.

4.7 .1.4.2 Low (operating and non-operating). This test shall be performed as specified in Procedures I and 11, Method 502 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.5 Humidity (non-operating) This test shall be Performed in accordance with Procedure III, Method 507 of MIL-STD-810 except that in Step 7, checkout shall be performed at the conclusion of the test only, with the item at standard ambient conditions. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.5.

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment and visual examination for compliance with 3.4.

4.7.3 Performance. The unit under test shall be allowed 2 minutes warm-up time prior to testing, except that 15 minutes is permitted for the low temperature tests of 4.7.1.4.2. A1l test equipment shall be allowed 15 minutes warm-up time prior to testing. The selected receiver-transmitters shall be tested as follows:

4.7.3.1 Interlock switch. Using standard measuring equipment, verify that the interlock switch fulfills the continuity requirements of 3.5.1.

4.7.3.2 Clutter lock-on switch. Using standard measuring equipment, verify that the clutter lock-on switch fulfills the continuity requirements **of** 3.5.2.

4.7.3.3 overload reset switch. Using standard measuring equipment, verify that the overload reset switch fulfills the continuity requirements of 3.5.3.

4.7.3.4 Indicator lamps.

4.7.3.4.1 Modulator overload indicator. Using the test setup in Figure 2, verify that the modulator overload indicator meets the requirements of 3.5.4.1.

4.7.3.4.2 High voltage power supply overload indicator. Using the test setup shown in Figure 2, verify that the high voltage power supply overload indicator meets the requirements of 3.5.4.2.

4.7.3.4.2 Power indicator lamp.

a. Connect the receiver-transmitter in the test setup shown in Figure 3. Verify that all power supplies and all test switches are turned to OFF.

b. Turn on power supply (item 13, Figure 3) and adjust for 24 V output.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:41Z
Check the source to verify that this is the current version before use.

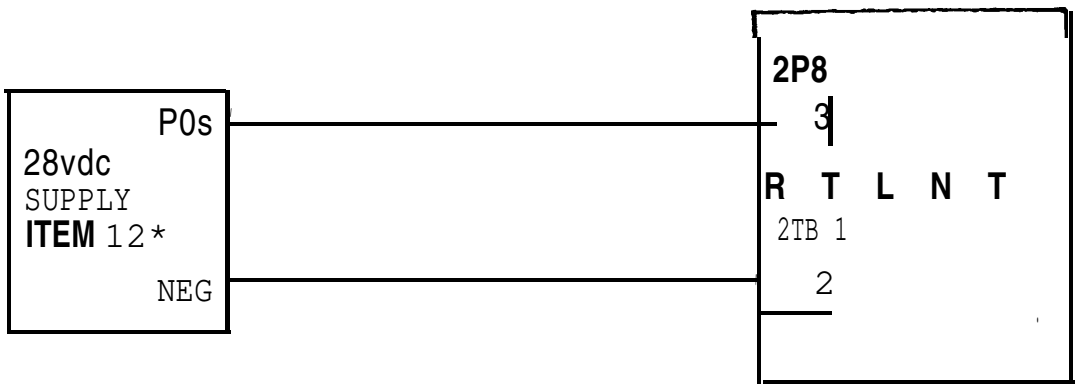


FIGURE 2. Test setup.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:41Z
Check the source to verify that this is the current version before use.

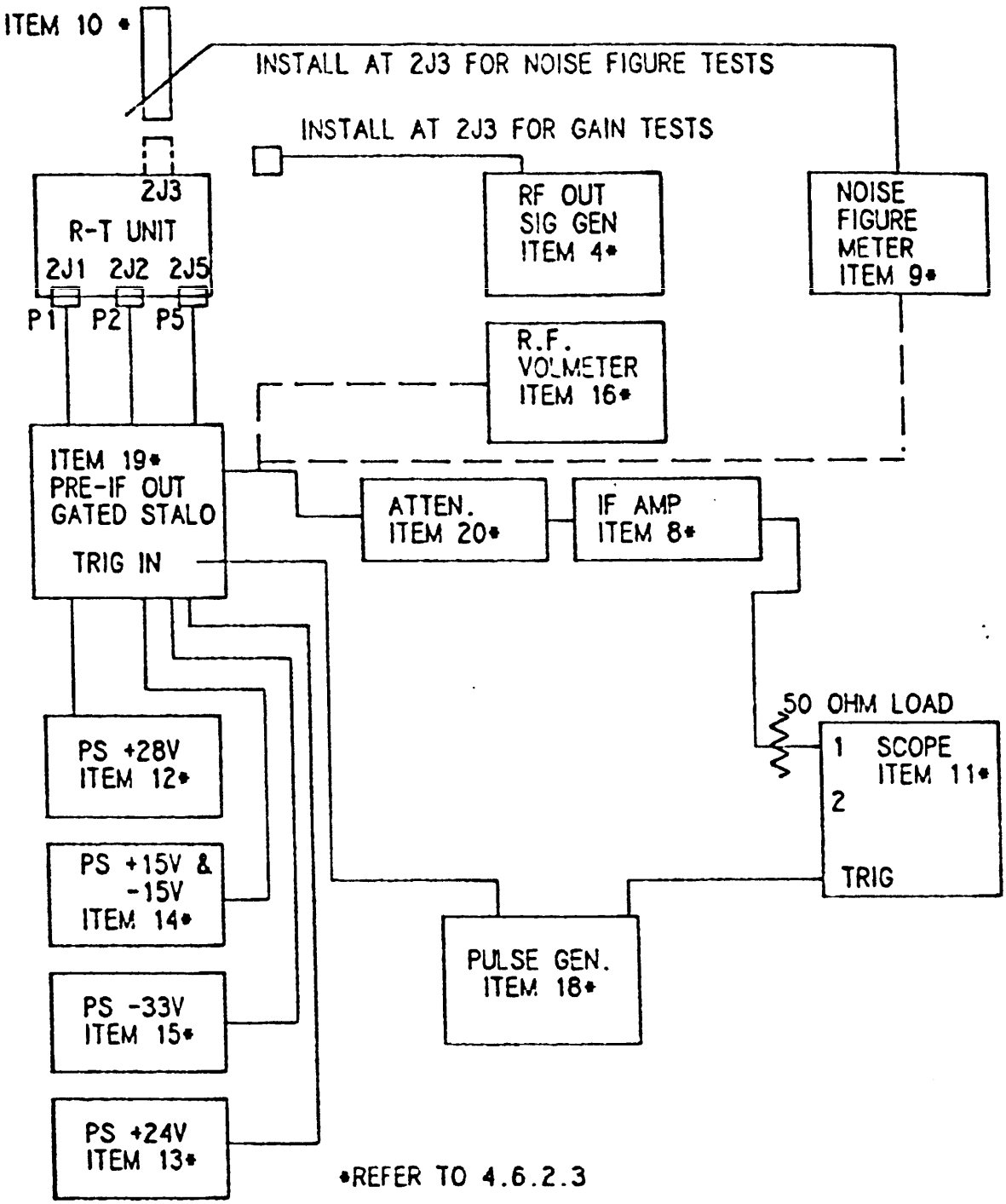


FIGURE 3. Test setup for standby operation.

Source: https://assist.dla.mil -- Downloaded: 202610-23T03:41Z
Check the source to verify that this is the current version before use.

c. Place the interlock switch in the manual override position. Place test fixture switch S1 to the ON position. The power indicator lamp shall illuminate.

4.7.3.5 Standby operation.

4.7.3.5.1 Blower operation. With the equipment operated as in 4.7.3.4.3, the blower shall operate and provide an intake airflow.

a. Adjust test equipment, shown in Figure 3, as required in Table V (Test equipment adjustments).

TABLE V. Test equipment adjustments.

TEST EQUIPMENT	ADJUST FOR
Pulse generator, item 18	20 kHz PRF, pulse width 0.6 ± 0.1 us, pulse amplitude +15 V peak
Power supply, item 12	+28 vdc
Power supply, item 14	Output no. 1 for +15 vdc Output no. 2 for -15 Vdc
Power supply, item 15	-33 vdc
Power supply, item 14	+24 Vdc
Other items	As required by test procedure

4.7.3.5.2 Gated stalo output.

a. Turn on all power supplies and connect a 50 ohm coaxial test cable between gated stalo output on test fixture and charnel 2 input on the oscilloscope (item 11, Figure 3). Terminate the cable with a 50 ohm load at the oscilloscope. Turn on test fixture switch S2.

b. Using the delayed sweep feature and external trigger from-the pulse-generator (item 18, Figure 3), adjust the oscilloscope for a 1 microsec time base. Observe that the gated stalo output meets the requirements of 3.5.5.2. Disconnect the test cable and reconnect 2P23 to 2A1J2.

4.7.3.5.3 Pulse shaper output.

a. Disconnect 2P23 from 2A1J2 and connect a coaxial test cable between 2P23 and charnel 2 input on the oscilloscope (item 11, Figure 3).

b. Verify that the pulse shaper output pulse at 2P23 meets the requirements of 3.5.5.3. Disconnect the test cable.

Source: https://assist.dla.mil - Downloaded: 2026-10-23T03:41Z
Check the source to verify that this is the current version before use.

4.7 .3.5.4 Overall gain.

a. Connect the signal generator (item 4, Figure 3) output to 2J3 on the RT unit and set the output to -40 DBM.

b. DiSCOnnect 2P19 from 2A1A5J2 and connect the RF voltmeter (item 16, Figure 3) to 2AlA5J2.

c. Plug the channel 4 crystal module into the microwave assembly and tune the-signal generator for a peak indication on the RF voltmeter. Set the AGC voltage control, R1 on the test fixture, for 0.0 V AGC. The RF voltmeter shall indicate not less than -23 DBM and not more than -18 DBM.

4.7.3.5.5 AGC action.

a. Disconnect the RF voltmeter and connect 2A1A5J2 to the attenuator (item 20, Figure 3), i-f amplifier (item 8, Figure 3) and oscilloscope.

b. Set 20 DB attenuation in the attenuator and establish a reference level on the oscilloscope, peaking up the x-band signal generator tuning if necessary.

c. Advance the AGC voltage, using pot R1 on the test fixture, from 0.0 to +6.0 V, at the same time reducing the attenuator to maintain reference level established in b. At 6.0 V AGC and O DB attenuation, the signal output shall not exceed the reference level.

4.7.3.5.6 Noise figure.

a. Connect the x-band noise source waveguide (item 10, Figure 3) to 2J3 and connect the noise figure meter (item 9, Figure 3) to the source.

b. Connect a 50-ohm coaxial test cable from 2A1A5J2 to the i-f input on the noise figure meter.

c. Measure noise figure using the procedure given in the manufacturer's manual for the noise figure meter. Reading shall not be more than 12 DB.

d. Repeat c, installing each of the other five crystal modules, in turn, in the microwave assembly. The reading on each channel shall not be more than 12 DB. Disconnect the noise figure test equipment and reconnect 2P19 at 2A1A5J2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:41Z
Check the source to verify that this is the current version before use.

4.7.3.5.7 Switch drive pulse. Connect a test cable to 2A1A7J1 and 2P5, using a tee connector, and connect the other end of the cable to the oscilloscope charnel 2 (item 11, Figure 3). The switch drive pulse parameters shall meet the requirements of 3.5.5.7. Remove the test cable and the connector and reconnect 2P5 to 2A1A7J2.

4.7.3.5.8 Diode switch isolation.

a. Connect the x-band signal generator to 2J3. Connect a test cable from pre-amp i-f output connector on the test fixture to channel 2 input on the oscilloscope with a 50 ohm load on the oscilloscope end.

b. Set the x-band generator output level to -20 DBM and peak the frequency for maximum deflection on the oscilloscope.

c. Adjust the oscilloscope delayed sweep for a 5 microsec time base synchronous with pulse generator trigger (item 18, Figure 3).

d. With the display as shown in Figure 4, verify that the RF level during the gated off time is 45 DB or less than the level during the on time.

4.7.3.6 Radiate mode.

4.7.3.6.1 Average output power.

a. Connect the R-T unit in the test setup as shown in Figure 5.

b. Turn on switch S3 of the test fixture and tune the klystron for maximum output at the operating frequency. The power meter (item 7, Figure 5) shall indicate not less than 0.426 mW. (Note: With 40 DB coupler, 0.426 mW at the 40 DB port indicates 4.26 W at through port.)

c. Repeat b on each of the five other channels. Turn off test fixture switch s3.

4.7.3.6.2 Transmit pulse timing.

a. Connect a coaxial test cable between 2J6 and the RF attenuator (item 20, Figure 5). Disconnect 2P33 from 2DC3 and connect the output of the attenuator to 2P33 in the receivertransmitter unit. Set the attenuator for 14 DB attenuation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:41Z
Check the source to verify that this is the current version before use.

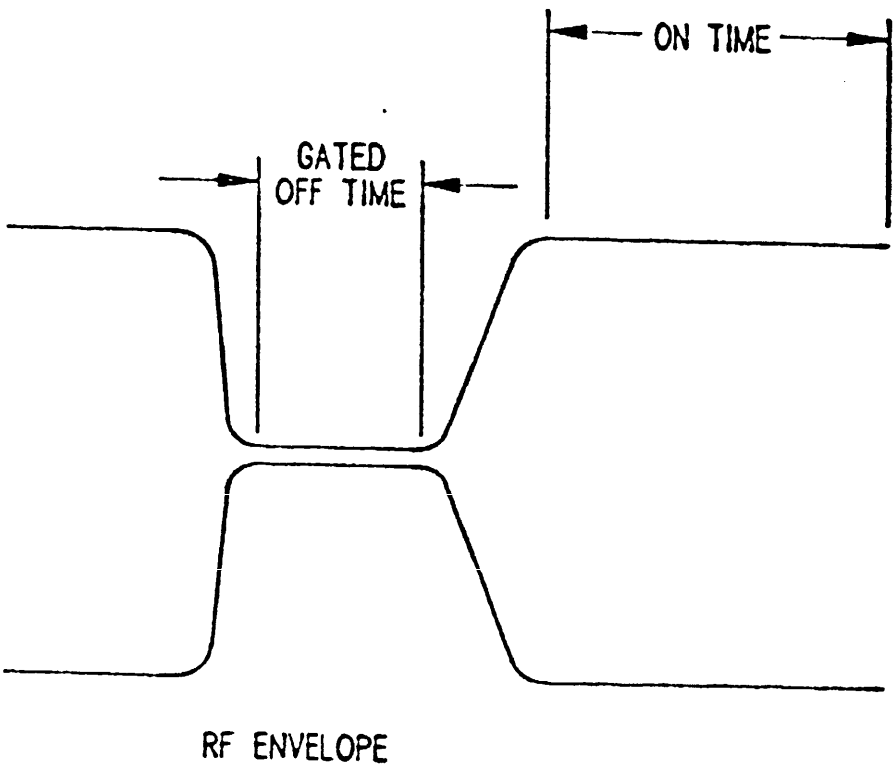


FIGURE 4. Diode switch isolation pattern.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:41Z
Check the source to verify that this is the current version before use.

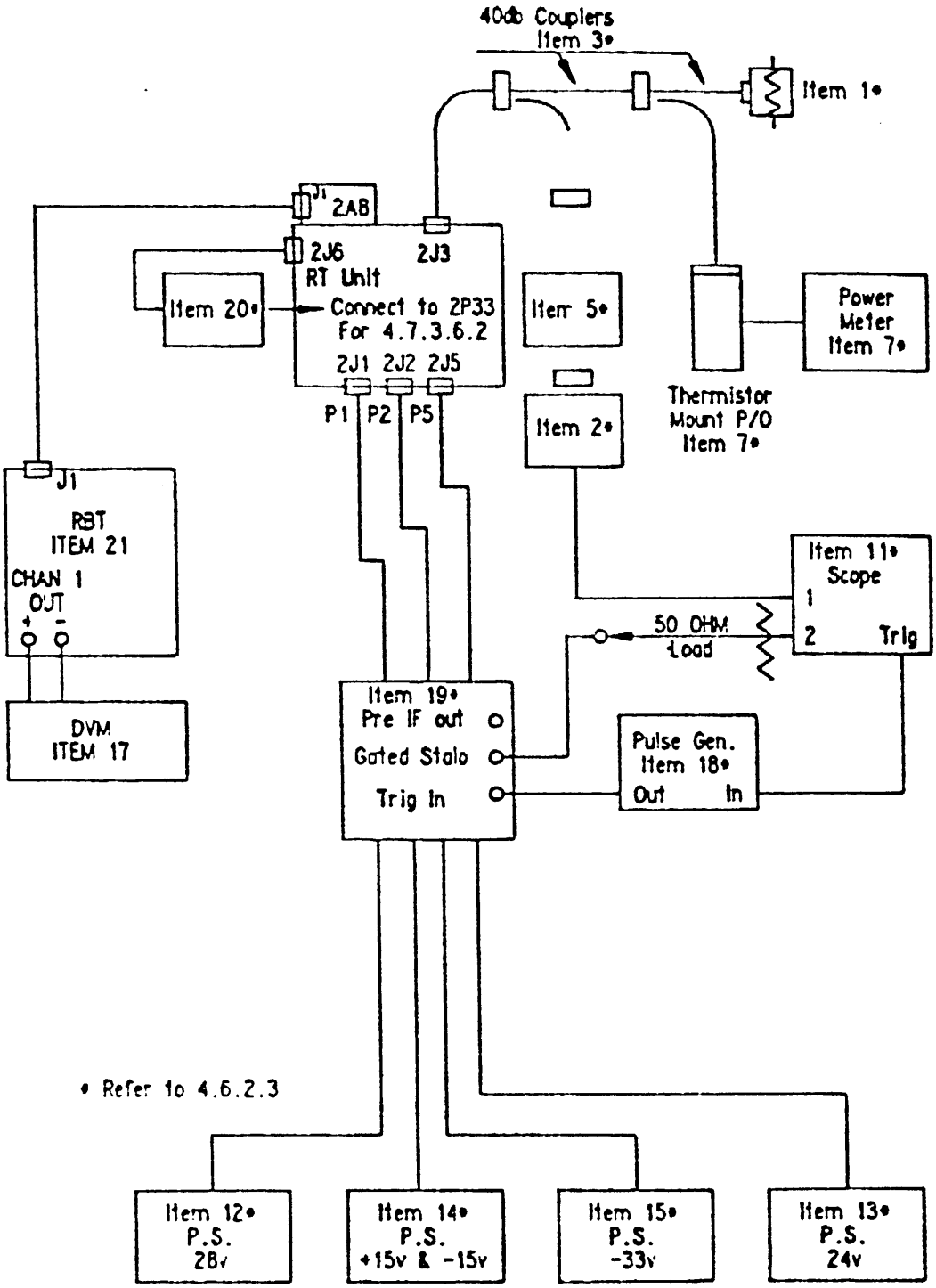


FIGURE 5. Test setup for radiate mode.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:41Z
Check the source to verify that this is the current version before use.

b. Turn on test fixture switch S3. With the oscilloscope channel selector on 'alternate" and a 1 **microsec time** base, adjust the delayed sweep to view the detected RF pulse and the gated stalo burst. The timing requirements of 3.5.6.2 shall be met. Turn off test fixture S3 and disconnect the attenuator from 2J6. Restore 2P33 to its original connection.

4.7.3.6.3 Transmit pulse envelope.

a. Turn on test fixture S3. Observe that the detected RF envelope on charnel 1 of the oscilloscope meets the requirements of 3.5.6.3. . Turn off switch S3.

4.7.3.6.4 Transmit frequency.

a. Turn on 2S6, service radiate switch. While observing the transmit pulse display, tune the cavity frequency meter (item 5, Figure 5) for a slight dip in pulse amplitude. At the point of maximum dip, read the frequency from the scale of the frequency meter. The reading shall meet the requirements of 3.5.6.4.

b. Repeat **a** with each of the other five crystal modules installed, tuning the klystron to each operating frequency.

4.7.3.7 Subclutter visibility test. After performing the tests of 4.7.3 through 4.7.3.6.4b, substitute the receiver-transmitter into a Radar Set AN/VPS-2A and perform the subclutter visibility (SCV) test as specified in MIL-R-70560. The Radar Set **AN/VPS-2A and test facilities used shall be as specified** in **MIL-R-70560 .**

4.7.4 Remote BIT assembly. The remote BIT assembly shall be tested using a digital voltmeter (item 17, Table IV) and a remote BIT tester (9361217) or equivalent to verify conformance with Table VI (BIT circuit card test data) and MIL-C-70574. Set up test equipment **per** Figure 5.

TABLE VI. BIT circuit card test data.

STATION 1234 CODE	CHANNEL NAME	VOLTAGE OUT
8 0 0 0	Switch Driver Output	1.33 ± 0.25 Vdc
8 1 0 1	Pulse Shaper Output	8.20 ± 0.25 vdc
8 2 0 2	Internal Cal (-15 sat)	0.00 ± 0.01 Vdc

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:41Z
Check the source to verify that this is the current version before use.

TABLE VI. BIT circuit card test data (continued].

STATION 1234 CODE	CHANNEL NAME	VOLTAGE OUT
8 3 0 3	Internal Cal (+15 sat)	10.00 ± **0.05** Vdc
8 4 0 4	60 MHz Oscillator	Greater than 7.0 Vdc
8 5 0 5	Inter Cal (+15 V)	6.10 ± 0.30 Vdc
8 7 0 7	Transmit Gate Output	Greater than 5.5 Vdc
8 8 0 8	Radar Operate Control	9.04 ± 2.5 vdc
8 10 8 10	Inter Cal (-15 V)	3.85 ± 0.20 Vdc
8 11 0 11	**+28** V Standby	9.04 ± 0.25 Vdc
8 15 0 15	Internal Cal GND	5.oo ± 0.03 Vdc

4.8 Soldering. Soldering Processes shall be inspected by certified inspectors to the requirements and acceptance criteria of Note 2 of Drawing 12950877 as specified in paragraph 3.7.

5. PACKAGING

5.1 General. Preservation, packing and marking shall be in accordance with Special Packaging Instructions PS9361152. The level of protection shall be as specified in the procurement document.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Receiver-Transmitter, Radar, RT-860A/VPS-2A is part of Radar Set AN/VPS-2A. The receiver-transmitter provides sufficient pulsed RF power to obtain usable echo return from targets of interest and converts these target signals to intermediate frequency for use in other units of the system.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:41Z
Check the source to verify that this is the current version before use.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Issue-of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).
c. Applicable stock number.
d. **First** article sample requirements.
e. Packaging requirements, if other than specified in Section 5.
f. Serialization requirements, if applicable.
g. Certification of conformance for each lot of shipment of product.

6.3 Submission of contractor inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army AMCCOM, Attn: SMCAR-QAF-I, Picatinny Arsenal, NJ **07806-5000.** This address will be specified on-the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC .

6.5 Subject term (Key word) listing.

Built-in-test (BIT)
Fire Control System
Product Improved Vulcan Air Defense System (PIVADS)

6.6 Changes from previous issue. The previous issues were for ARDEC internal use Only. The were not forwarded to Naval Publication and Forms Center (NPFC) for publication. This issue is initial public release for unlimited distribution.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1285-A742)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:41Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-R-70561	24 JANUARY 1992

3. DOCUMENT TITLE
RECEIVER-TRANSMITTER, RADAR SET RT-860A/VPS-2A: 9361152

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)
U.S ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial: (201) 724-6674 (2) AUTOVON: 880-6674
c. ADDRESS (Include Zip Code) ATTN: SMACA-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

DD Form 1426, OCT 89 *Previous editions are obsolete.* 198/290

Source: https://assist.dla.mil -- Downloaded: 202610-23T03:41Z
Check the source to verify that this is the current version before use.